Bias Collapse in Shallow ReLU Networks

Numerical Evidence for Open Problems 4.1, 4.2, and 4.3

NSF RTG Project 4

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Motivation

The overparameterization puzzle:

- Modern networks have far more parameters than the problem demands
- Classical learning theory predicts overfitting; practice shows robust generalization
- Why does the optimizer self-simplify?

Bias collapse: a concrete mechanism

- Gradient flow automatically eliminates redundant neurons — no regularization required
- $m = 1500$ neurons $\longrightarrow$ effective complexity of $k = 7$ neurons
- The number of surviving *bias clusters* equals the number of *inflection points* of the target function

This work

Rigorous numerical investigation of three open problems:

1. **OP 4.1:** Does $C(m, f^*) \to k$ as $m \to \infty$?
2. **OP 4.2:** Does collapse generalize to $d \geq 2$ dimensions?
3. **OP 4.3:** Is there a provable L^2 pruning bound after collapse?

78 runs, 9 targets, m up to 5000.

Model and Gradient Flow

1D shallow ReLU network:

$$f(x) = \sum_{j=1}^{m} a_j\,\sigma(x - b_j), \quad \sigma = \max(0,\cdot), \quad x \in [-1,1]$$

Each neuron places a *kink* at bias b_j with slope change a_j.

Gradient flow on MSE $\mathcal{L} = \frac{1}{2}\int_{-1}^{1}(f - f^*)^2\,dx$**:**

$$\dot a_j = -\int_{-1}^{1}(f - f^*)\,\sigma(x - b_j)\,dx$$

$$\dot b_j = a_j \int_{b_j}^{1}(f - f^*)\,dx$$

Solved via `scipy` RK45, $T = 500$, rtol $= 10^{-4}$.
Stationarity criterion: $\max_j |\dot a_j|, |\dot b_j| < 0.01$.

Key measured quantity:

$$C(m, f^*) \;=\; \#\{\text{bias clusters}\}$$

at stationarity; gap threshold 0.02.

Fixed-point condition

Each *active* neuron ($a_j \neq 0$) satisfies $R_j := \int_{b_j}^{1}(f - f^*)\,dx \approx 0$ at stationarity. Verified numerically at every converged run; connects to the theory of OP 4.1.

The Three Open Problems

OP 4.1 – Cluster Count

$$\lim_{m \to \infty} C(m, f^*) = k$$

$k =$ number of sign-changing inflection points of f^*

Sub-problems:
4.1.1: upper bound on C
4.1.4: which neurons survive?

OP 4.2 – Higher Dimensions

For $f^* : [-1, 1]^2 \to \mathbb{R}$, does gradient flow concentrate the neuron measure onto a lower-dimensional set in hyperplane parameter space?

2D network:
$f(x) = \sum_j a_j \, \sigma(w_j^\top x + b_j)$

Collapse modes depend on the structure of f^* (ridge, separable, radial).

OP 4.3 – Pruning Bound

After collapse, prune to k neurons with provable L^2 error:

$$\left\| \tilde{f} - f \right\|_{L^2} \leq \delta \cdot \sum_{j=1}^{m} |a_j|$$

$\delta =$ max intra-cluster diameter
$\tilde{f} =$ merged k-neuron network

Connects collapse to formal approximation theory.

Experimental Design

Nine target functions:

$f^*(x)$	k	type
$\sin(\pi x)$	1	trig
x^3	1	poly
$\sin(2\pi x)$	3	trig
$x^5 - 3x^3$	3	poly
$\sin(3\pi x)$	5	trig
$\sin(4\pi x)$	7	trig
$\sin(5\pi x)$	9	trig
$\sin(6\pi x)$	11	trig
$\sin(7\pi x)$	13	trig

Trig: $k = 2n - 1$. Poly: irregular inflection spacing.

Main sweep (`simulate_parallel.py`):

- $m \in \{50, 100, 250, 500, 1000, 2000, 3500, 5000\}$, $T = 500$
- **78 completed runs**, 9 targets; 76/78 fully stationary ($\max |\dot{a}_j| < 0.01$)

Supporting experiments:

Experiment	Runs	Purpose
`simulate_discrete.py`	52	Discrete GD vs. ODE
`lottery_ticket_experiment.py`	52	OP 4.1.4 (survival)
`verify_pruning.py`	78	OP 4.3 bound check
`collapse_v2.py`	9	OP 4.2, 2D targets
`instability_test.py`	22	Stability near k

OP 4.1 – The Rise-Then-Fall Dynamic

Universal pattern across all 9 targets:

1. **Rise:** Small m cannot populate all k inflection attractors simultaneously. C grows above k.
2. **Peak:** C reaches a maximum scaling with k (up to $C = 40$ for $k = 11$).
3. **Fall:** With sufficient redundancy, collapse proceeds and C decreases toward k.

Key observation

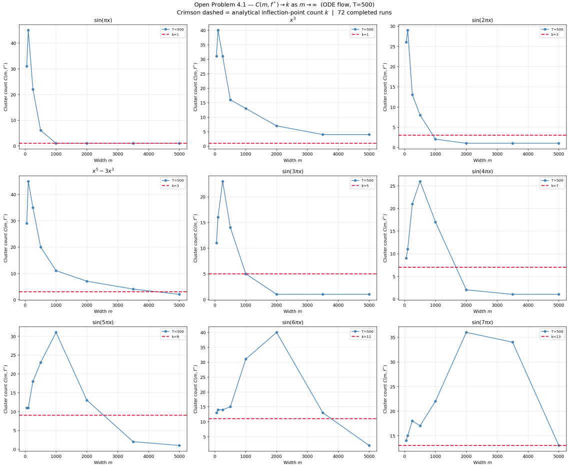
The peak of $C(m)$ shifts to **larger** m as k grows. High-k targets require much wider networks before the fall phase begins.

Selected $C(m)$ at $T = 500$:

Target	k	m=1000	m=2000	m=5000
$\sin(\pi x)$	1	**1**	1	1
$\sin(3\pi x)$	5	**5**	†1	†1
$\sin(4\pi x)$	7	17	2	†1
$\sin(7\pi x)$	13	22	36	‡**13**
x^5-3x^3	3	11	7	2
x^3	1	13	7	4

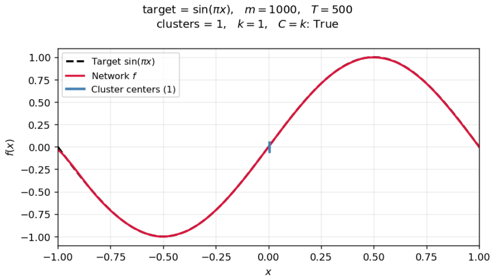
† Dense-packing artifact (biases uniformly spread).
‡ $C = k$ but not fully stationary $(\max|\dot{a}_j| = 0.034)$.

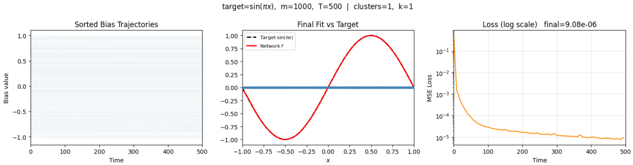
OP 4.1 – $C(m, f^*)$ Across All 9 Targets



$C(m, f^*)$ vs. m at $T = 500$ (78 runs, 9 targets). **Crimson dashed line** $= k$. All panels show rise-then-fall. Low-k trig targets reach $C = k$; $\sin(7\pi x)$ ($k = 13$) reaches $C = 13$ at $m = 5000$ for the first time (stationarity unconfirmed).

OP 4.1 – Confirmation: $\sin(\pi x)$, $k = 1$





$\sin(\pi x)$, $\quad m = 1000$, $\quad k = 1$

Top: Clean final fit vs. target. Blue ticks = single cluster center.

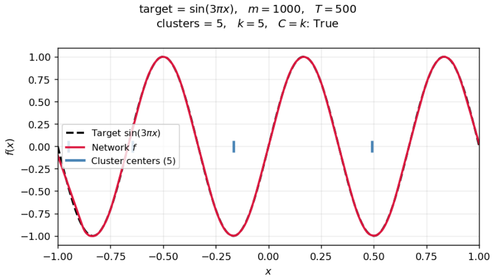
Bottom: Bias trajectories (all 1000 collapse to one location), final fit, and MSE loss on log scale.

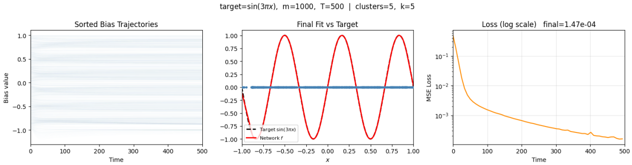
Strongest confirmation

$C = 1 = k$ for *all* $m \geq 1000$ through $m = 5000$, fully stationary $(\max |\dot{a}_j| < 3.6 \times 10^{-3})$.

Only target with $C = k$ confirmed robustly across the full m sweep.

OP 4.1 – Confirmation: $\sin(3\pi x)$, $k = 5$





$\sin(3\pi x)$, $\quad m = 1000$, $\quad k = 5$

Top: Clean final fit. Five blue ticks mark the 5 bias clusters, one per inflection point.

Bottom: Bias trajectories, final fit, MSE loss on log scale.

Non-monotone path to k

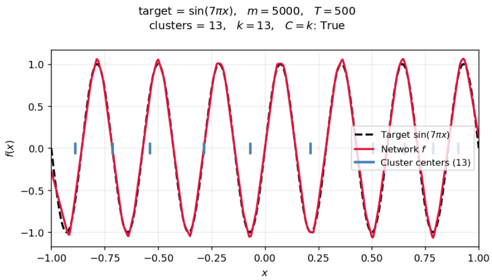
$$\underbrace{8}_{m=50} \to \underbrace{14}_{\text{peak},\ m=500} \to \underbrace{\mathbf{5}}_{k,\ m=1000} \to \underbrace{\dagger 1}_{m=2000}$$

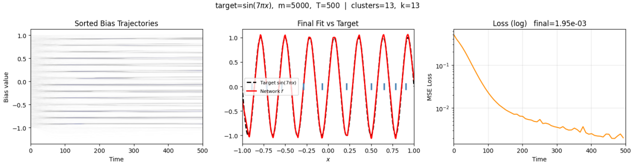
Rises above k, falls through $k = 5$, then †dense-packing artifact.

Opens a question

Why does C overshoot and fall below k at larger m? (next slide)

OP 4.1 – New at $m = 5000$: $\sin(7\pi x)$, $k = 13$





$\sin(7\pi x)$, $\quad m = 5000$, $\quad k = 13$

Top: Clean final fit. Thirteen blue ticks align with all 13 inflection points.

Most complex target tested. First time $C = k = 13$ is reached.

Result

$C = 13 = k$ at $m = 5000$.

Path: $14 \to 22 \to 36 \to 34 \to \mathbf{13}$

$(m = 50, 1000, 2000, 3500, 5000)$

Caveat

$\max |\dot{a}_j| = 0.034 > 0.01$. May be **transient** — longer T needed to confirm.

OP 4.1 – The Below-k Phenomenon

Three simulate_parallel runs reach $C < k$ at verified stationary states:

Target	k	m	C	$\max\lvert\dot{b}_j\rvert$
$\sin(2\pi x)$	3	1000	2	1.8×10^{-4}
$\sin(4\pi x)$	7	2000	2	4.5×10^{-4}
x^5-3x^3	3	5000	2	1.2×10^{-3}

All three are **genuinely stationary**. All report $C = 2$ regardless of k.

Interpretation A: True fixed points

Below-k states are genuine fixed points of the ODE at large m. The conjecture as stated is *false* for $k \geq 3$.

Interpretation B: Local minima

Gradient flow has multiple basins. Below-k states are initialization-dependent. The conjecture holds from the "right" initial condition.

Resolution (pending)

Re-run 3 cases with 2 additional seeds. $C = 2$ across all seeds $\Rightarrow$ A. C varies $\Rightarrow$ B.

OP 4.1 – Polynomial vs. Trigonometric Targets

Same k, very different convergence speed:

Target	k	m=1000	m=2000	m=5000
$\sin(\pi x)$	1	**1**	1	1
x^3	1	13	7	4
$\sin(2\pi x)$	3	2	†1	†1
$x^5 - 3x^3$	3	11	7	2

x^3 reaches only $C = 4$ at $m = 5000$, despite $k = 1$.

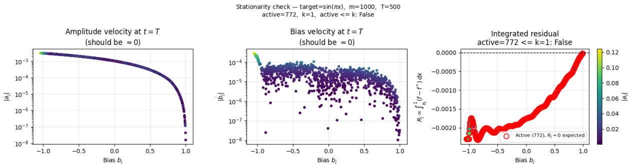
Why inflection geometry matters

For $\sin(n\pi x)$, the k inflection points are *evenly spaced* across $[-1, 1]$ — strong, symmetric attractors that neurons organize around early in training.

Polynomial targets have *irregular* spacing: weaker, asymmetric attractors. Neurons take far longer to find them.

Implication: a proof of OP 4.1 must account for attractor geometry, not just count.

OP 4.1 – Stationarity and Fixed-Point Verification



$\sin(\pi x)$, $m = 1000$, $T = 500$

Final (a_j, b_j) substituted back into the ODEs.

Left: $|\dot{a}_j| \approx 10^{-3}$–$10^{-5}$

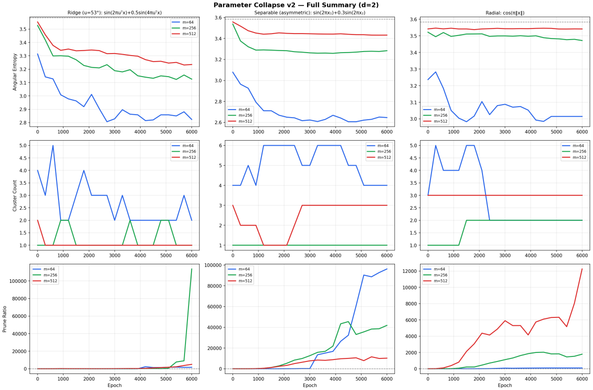
Center: $|\dot{b}_j| \approx 10^{-4}$–$10^{-5}$

Right: $R_j \approx 0$ for active neurons (red) — the theoretical fixed-point condition, satisfied numerically.

Discrete GD comparison

Discrete GD does *not* produce $C \to k$ for any target except $\sin(\pi x)$. The conjecture is specific to continuous ODE gradient flow dynamics.

OP 4.2 – Higher-Dimensional Collapse



Three 2D targets ($x \in [-1,1]^2$):

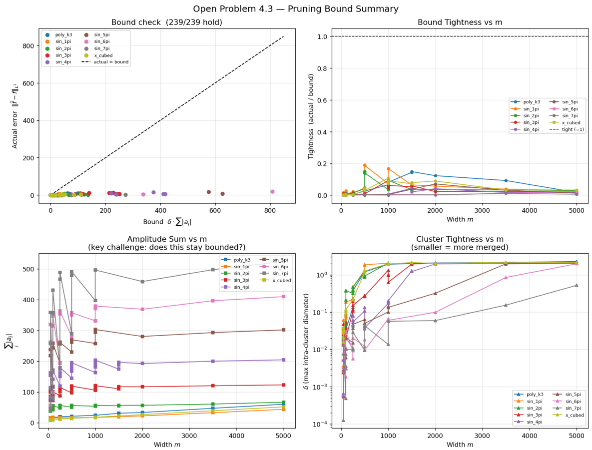
- **Ridge:** $\sin(2\pi u^\top x) + 0.5\sin(4\pi u^\top x)$
  Predicted: collapse to one direction
- **Separable:** $\sin(2\pi x_1) + 0.3\sin(2\pi x_2)$
  Predicted: two direction families
- **Radial:** $\cos(\pi\|x\|)$
  Predicted: no collapse (symmetry)

Key findings

Collapse **mode** tracks target structure at small m: ridge concentrates, separable splits, radial stays near uniform (entropy 3.54 vs. max 3.58).

Collapse **weakens** as m grows — opposite of the large-m theoretical prediction. Large networks fit the target without collapsing.

OP 4.3 – Pruning Bound Verification



Bound: $\left\|\tilde{f} - f\right\|_{L^2} \leq \delta \cdot \sum_j |a_j|$

78 / 78 runs: bound holds

Tightness (actual / bound): 0.0002 to 0.148. Substantial slack in all cases.

Key obstacle:

- $\sum_j |a_j| \sim \sqrt{m}$ empirically
- δ does not shrink with m at $T = 500$
- Bound loosens $\sim \sqrt{m}$ while actual error plateaus

Path to non-vacuousness

$\delta \to 0$ requires $T \to \infty$, *not* $m \to \infty$. Study $\delta(T)$ at fixed m to make the bound formally useful.

OP 4.1.4 – Lottery Ticket: Which Neurons Survive?

Setup: At $t = 0$, amplitudes $a_j \sim \mathcal{N}(0, 0.01)$ are random. Only the bias placement b_j is structured. Neurons near an inflection point are *geometrically lucky*.

Five conditions per (target, m): `full`, `geometric`, `bias_only`, `amp_only`, `random_k`

52 runs; 4 targets, $m \in \{500, 1000, 1500\}$.

Claim 2 (performance): FALSE

Lucky k neurons alone achieve $100\times$–$50{,}000\times$ higher loss than the full network. k kinks cannot approximate a smooth target; full redundancy is necessary.

Claim 1 (survival): bias position is key

geometric $\approx$ **bias_only** in every case (loss diff < 0.002). Initial amplitudes carry *no information*.

amp_only is the worst condition: good amplitudes with random biases performs worse than good biases with random amplitudes.

Conclusion:
Bias position at $t = 0$ is the sole informative quantity for collapse. The "winning ticket" is a good bias placement, not a good amplitude.

Instability Near k – Framework and Preliminary Results

Setup: Runs with $|C-k| \leq 2$ qualify.
22 qualifying runs: 8 exact_k, 8 above_k, 6 below_k.

Category	Test
exact_k ($C=k$)	Inject one extra neuron. Does it dissolve?
above_k ($k < C \leq k+2$)	Continue flow naturally. Does C fall to k?
below_k ($k-2 \leq C < k$)	Inject one neuron. Does C rise to k?

Preliminary finding

above_k "natural dissolution" tests terminate at $t \approx 0$ — above-k states are already ODE fixed points. They do **not** spontaneously dissolve.

Implication for OP 4.1.1

A Lyapunov proof that $C \leq k$ requires $C > k$ states to be unstable. These results suggest they are *metastable* — stable under small perturbations. The injection tests will determine large-perturbation stability.

Full results: in progress ($T_{\text{perturb}} = 1000$).

Summary of Evidence

Problem	Main finding	Status
OP 4.1: $C \to k$	Confirmed for $\sin(\pi x)$ ($k = 1$), all $m \geq 1000$. $\sin(7\pi x)$ reaches $C = 13 = k$ at $m = 5000$ (stationarity unconfirmed). Below-k at 4 targets (genuinely stationary).	**Supported** for $k = 1$; **open** for $k \geq 3$
OP 4.1.4: Lottery ticket	Bias position at $t = 0$ is sole informative quantity. Amplitude values irrelevant. Performance claim false.	**Resolved**
OP 4.2: 2D collapse	Collapse mode tracks target structure at small m. Weakens at large m (unexpected).	**Partial**
OP 4.3: Pruning bound	78 / 78 hold. $\sum \lvert a_j \rvert \sim \sqrt{m}$. Non-vacuousness requires $\delta \to 0$ as $T \to \infty$.	**Verified numerically**; **open theoretically**
Instability near k	above_k states are ODE fixed points; no spontaneous dissolution. Injection results pending.	**In progress**

Open Questions and Next Steps

OP 4.1 — open:

- **Below-k mystery:** True fixed points or initialization-dependent local minima?
  Next: 6 runs with additional seeds (3 cases)
- **4.1.1:** above_k states are metastable — a Lyapunov proof needs a refined energy function
- $\sin(7\pi x)$**:** Re-run at $T = 1000$ to confirm stationarity of $C = 13$

OP 4.2 — open:

- Why does collapse weaken at large m? Theory predicts the opposite.

OP 4.3 — path to a formal proof:

- Study $\delta(T)$ at fixed m: does $\delta \to 0$ as $T \to \infty$?
- Exponent α in $\sum_j |a_j| = O(m^\alpha)$; data suggests $\alpha \approx 1/2$
- If $\delta = O(T^{-\gamma})$ the bound becomes non-vacuous as $T \to \infty$

Immediate priorities

(1) Multiple seeds for below-k cases
(2) Instability test final results
(3) $\delta(T)$ trajectory study
(4) $\sin(7\pi x)$ at $T = 1000$

Conclusion

What we showed:

- $C \to k$ confirmed for $\sin(\pi x)$ ($k = 1$), $m \geq 1000$. First high-k evidence: $\sin(7\pi x)$ reaches $C = 13$ at $m = 5000$.
- **Rise-then-fall** is universal: peak scales with k; inflection geometry sets the convergence rate.
- **Below-k:** 4 targets have $C < k$ at genuinely stationary states — open challenge for OP 4.1.
- **OP 4.3:** Bound holds 78/78; $\delta \to 0$ with T is the path to a formal proof.
- **Lottery ticket:** bias position at $t = 0$ is the sole informative quantity.

Broader implication

Gradient flow is not only an optimizer — it is a **compression algorithm** that discovers the minimum-complexity representation of the target's intrinsic structure.

Inflection points of f^* act as **fixed-point attractors**: neurons near one survive; the rest dissolve.

A rigorous proof would explain generalization in overparameterized networks.

Thank you

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